Hadith Hallucination Benchmark — Results Tables

Table 1. Overall Summary (mean across all six Hadith collections)

3,000 responses per model–language cell (500 × 6 collections). All metrics used in the evaluation protocol are included: BLEU (lexical), cosine similarity (semantic), and the full 5-way classification percentages.

Model	Lang	Avg BLEU	Avg Cosine	%Correct	%Theme-only	%Partial	%Halluc.	%Refused
ALLaM-7B	Arabic	0.015	0.480	0.00	4.37	41.37	54.27	0.00
ALLaM-7B	English	0.006	0.518	0.00	0.27	59.30	40.43	0.00
Gemma3-4B	Arabic	0.019	0.736	0.00	39.83	50.80	9.37	0.00
Gemma3-4B	English	0.008	0.558	0.00	0.23	74.23	25.53	0.00
Jais-7B	Arabic	0.024	0.567	0.00	9.90	54.27	35.83	0.00
Jais-7B	English	0.007	0.535	0.00	0.03	66.10	33.87	0.00
Mistral-7B	Arabic	0.006	0.719	0.00	32.27	59.73	8.00	0.00
Mistral-7B	English	0.007	0.559	0.00	0.47	75.03	24.50	0.00
Phi4-mini-3.8B	Arabic	0.019	0.497	0.00	6.97	44.83	47.50	0.70
Phi4-mini-3.8B	English	0.005	0.447	0.00	0.07	41.13	58.80	0.00

Table 2. Mistral-7B — Per-Collection Breakdown

500 responses per collection–language cell. All metrics used in the evaluation protocol are included.

Collection	Lang	Avg BLEU	Avg Cosine	%Correct	%Theme-only	%Partial	%Halluc.	%Refused
Sahih Bukhari	English	0.006	0.538	0.00	0.20	66.20	33.60	0.00
Sahih Muslim	English	0.004	0.529	0.00	0.40	68.60	31.00	0.00
Sunan Abi Da'ud	English	0.007	0.567	0.00	0.00	76.80	23.20	0.00
Jami' al-Tirmidhi	English	0.010	0.570	0.00	0.60	77.40	22.00	0.00
Sunan an-Nasa'i	English	0.006	0.592	0.00	0.60	85.60	13.80	0.00
Sunan Ibn Majah	English	0.007	0.562	0.00	1.00	75.60	23.40	0.00
Sahih Bukhari	Arabic	0.007	0.728	0.00	34.40	59.60	6.00	0.00
Sahih Muslim	Arabic	0.004	0.733	0.00	34.20	58.00	7.80	0.00
Sunan Abi Da'ud	Arabic	0.005	0.717	0.00	35.00	57.20	7.80	0.00
Jami' al-Tirmidhi	Arabic	0.008	0.722	0.00	38.40	49.60	12.00	0.00
Sunan an-Nasa'i	Arabic	0.007	0.699	0.00	24.60	68.20	7.20	0.00
Sunan Ibn Majah	Arabic	0.006	0.715	0.00	27.00	65.80	7.20	0.00

Table 3. Gemma3-4B — Per-Collection Breakdown

500 responses per collection–language cell. All metrics used in the evaluation protocol are included.

Collection	Lang	Avg BLEU	Avg Cosine	%Correct	%Theme-only	%Partial	%Halluc.	%Refused
Sahih Bukhari	English	0.005	0.551	0.00	0.00	71.40	28.60	0.00
Sahih Muslim	English	0.012	0.540	0.00	0.60	69.60	29.80	0.00
Sunan Abi Da'ud	English	0.007	0.548	0.00	0.00	70.60	29.40	0.00
Jami' al-Tirmidhi	English	0.006	0.567	0.00	0.20	76.80	23.00	0.00
Sunan an-Nasa'i	English	0.010	0.579	0.00	0.40	81.20	18.40	0.00
Sunan Ibn Majah	English	0.007	0.564	0.00	0.20	75.80	24.00	0.00
Sahih Bukhari	Arabic	0.019	0.684	0.00	27.40	57.40	15.20	0.00
Sahih Muslim	Arabic	0.015	0.703	0.00	32.00	54.60	13.40	0.00
Sunan Abi Da'ud	Arabic	0.011	0.736	0.00	37.40	53.60	9.00	0.00
Jami' al-Tirmidhi	Arabic	0.027	0.774	0.00	53.40	39.00	7.60	0.00
Sunan an-Nasa'i	Arabic	0.013	0.746	0.00	41.20	51.20	7.60	0.00
Sunan Ibn Majah	Arabic	0.030	0.776	0.00	47.60	49.00	3.40	0.00

Table 4. Phi4-mini-3.8B — Per-Collection Breakdown

500 responses per collection–language cell. All metrics used in the evaluation protocol are included.

Collection	Lang	Avg BLEU	Avg Cosine	%Correct	%Theme-only	%Partial	%Halluc.	%Refused
Sahih Bukhari	English	0.004	0.546	0.00	0.20	68.80	31.00	0.00
Sahih Muslim	English	0.011	0.515	0.00	0.00	63.40	36.60	0.00
Sunan Abi Da'ud	English	0.004	0.383	0.00	0.00	21.40	78.60	0.00
Jami' al-Tirmidhi	English	0.004	0.400	0.00	0.00	28.60	71.40	0.00
Sunan an-Nasa'i	English	0.004	0.434	0.00	0.00	37.60	62.40	0.00
Sunan Ibn Majah	English	0.004	0.402	0.00	0.20	27.00	72.80	0.00
Sahih Bukhari	Arabic	0.015	0.536	0.00	8.00	54.00	37.00	1.00
Sahih Muslim	Arabic	0.020	0.492	0.00	3.60	49.20	47.20	0.00
Sunan Abi Da'ud	Arabic	0.013	0.447	0.00	5.80	36.20	58.00	0.00
Jami' al-Tirmidhi	Arabic	0.016	0.468	0.00	12.00	25.80	60.00	2.20
Sunan an-Nasa'i	Arabic	0.023	0.518	0.00	7.00	51.00	41.80	0.20
Sunan Ibn Majah	Arabic	0.027	0.518	0.00	5.40	52.80	41.00	0.80

Table 5. Jais-7B — Per-Collection Breakdown

500 responses per collection–language cell. All metrics used in the evaluation protocol are included.

Collection	Lang	Avg BLEU	Avg Cosine	%Correct	%Theme-only	%Partial	%Halluc.	%Refused
Sahih Bukhari	English	0.005	0.524	0.00	0.00	61.00	39.00	0.00
Sahih Muslim	English	0.010	0.523	0.00	0.00	65.40	34.60	0.00
Sunan Abi Da'ud	English	0.006	0.536	0.00	0.00	63.40	36.60	0.00
Jami' al-Tirmidhi	English	0.009	0.541	0.00	0.00	69.40	30.60	0.00
Sunan an-Nasa'i	English	0.007	0.560	0.00	0.20	73.20	26.60	0.00
Sunan Ibn Majah	English	0.007	0.527	0.00	0.00	64.20	35.80	0.00
Sahih Bukhari	Arabic	0.012	0.516	0.00	5.60	46.40	48.00	0.00
Sahih Muslim	Arabic	0.023	0.579	0.00	10.80	55.40	33.80	0.00
Sunan Abi Da'ud	Arabic	0.033	0.628	0.00	14.60	61.80	23.60	0.00
Jami' al-Tirmidhi	Arabic	0.009	0.483	0.00	5.60	37.40	57.00	0.00
Sunan an-Nasa'i	Arabic	0.017	0.563	0.00	7.00	60.00	33.00	0.00
Sunan Ibn Majah	Arabic	0.051	0.636	0.00	15.80	64.60	19.60	0.00

Table 6. ALLaM-7B — Per-Collection Breakdown

500 responses per collection–language cell. All metrics used in the evaluation protocol are included.

Collection	Lang	Avg BLEU	Avg Cosine	%Correct	%Theme-only	%Partial	%Halluc.	%Refused
Sahih Bukhari	English	0.006	0.517	0.00	0.20	59.20	40.60	0.00
Sahih Muslim	English	0.005	0.482	0.00	0.20	48.80	51.00	0.00
Sunan Abi Da'ud	English	0.006	0.527	0.00	0.20	61.20	38.60	0.00
Jami' al-Tirmidhi	English	0.006	0.525	0.00	0.00	62.60	37.40	0.00
Sunan an-Nasa'i	English	0.005	0.533	0.00	0.60	61.60	37.80	0.00
Sunan Ibn Majah	English	0.005	0.526	0.00	0.40	62.40	37.20	0.00
Sahih Bukhari	Arabic	0.021	0.517	0.00	4.00	49.40	46.60	0.00
Sahih Muslim	Arabic	0.015	0.482	0.00	4.40	41.40	54.20	0.00
Sunan Abi Da'ud	Arabic	0.011	0.473	0.00	2.40	45.60	52.00	0.00
Jami' al-Tirmidhi	Arabic	0.010	0.438	0.00	6.40	27.20	66.40	0.00
Sunan an-Nasa'i	Arabic	0.022	0.505	0.00	3.80	49.40	46.80	0.00
Sunan Ibn Majah	Arabic	0.013	0.465	0.00	5.20	35.20	59.60	0.00